

# Oleksii SOKOLIUK

## PERSONAL DATA

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DATE OF BIRTH: 3<sup>rd</sup> October 2005  
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## WORK EXPERIENCE

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| JUL–SEPT 2025  | <i>Visiting Student – Institute of Astronomy, University of Cambridge</i><br>Supervisor: Prof. John Webb, Projects: <ul style="list-style-type: none"><li>– A very generalised approach of modeling the metal content in the early and late universe</li><li>– Determining Instrumental Profile of the ESPRESSO instrument for Rigorous Tests of Fundamental Constants and Redshift Drift</li></ul> |
| AUG–SEPT 2024  | <i>Visiting Student – KICC, University of Cambridge</i><br>Supervisor: Prof. John Webb, Project: <ul style="list-style-type: none"><li>– The cosmic web's Lyman-<math>\alpha</math> glow at <math>z \approx 2.5</math>; varying hydrodynamic models, dust, and wide-field, narrow-band imaging detection</li></ul>                                                                                  |
| MAY 2021 – NOW | <i>Research Scholar – Main Astronomical Observatory, NAS of Ukraine</i><br>Supervisor: Prof. Iryna Vavilova                                                                                                                                                                                                                                                                                         |

## EDUCATION

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SEPT 2023 - MAY 2027 Bachelor of Science (Hons) in PHYSICS  
University of Aberdeen, United Kingdom  
JULY 2023 St. Joseph's College, Dumfries

## TALKS & POSTER PRESENTATIONS

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| [1] Institute of Astronomy, University of Cambridge<br>Wednesday Colloquia                                                                | 16 Jul 2025<br>(invited talk)              |
| [2] Astronomy Group, University of St. Andrews<br>Lunchtime Talks                                                                         | 1 Jul 2025<br>(invited talk)               |
| [3] Mullard Space Science Laboratory, University College London<br>UCL Astrophysics seminar                                               | 23 Jan 2025<br>(invited talk)              |
| [4] GRANDMA collaboration (a part of LIGO/VIRGO)<br>Seminar/Telecon                                                                       | 05 Dec 2024<br>(invited talk)              |
| [5] Institute for Computational Cosmology, Durham University<br>Friday Lunchtime Astrophysics Talks (FLAT)                                | 14 Jun 2024<br>(invited talk)              |
| [6] Kobe International Conference Center, Kobe Port Island<br>CCP2023 - 34th IUPAP Conference on Computational Physics                    | 4 Aug - 8 Aug 2023<br>(contributed talk)   |
| [7] Faculdade de Ciências da Universidade de Lisboa<br>CosmoVerse@Lisbon, First Annual Conference                                         | 30 May - 1 Jun 2023<br>(poster)            |
| [8] Faculty of Physics, Odesa I.I. Mechnikov National University<br>XXI Gamow International Astronomical Conference-School                | 16 Aug - 20 Aug 2021<br>(contributed talk) |
| [9] Faculty of Physics, Taras Shevchenko National University of Kyiv<br>27-th Young Scientists' Conference on Astronomy and Space Physics | 26 Apr - 30 Apr 2021<br>(contributed talk) |

## SELECTED PUBLICATIONS

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As of Sept 2025, 24 papers were published in international, peer reviewed journals with 14 papers published as a first author, 2 as a second author. In total, those papers have >730 citations and  $h$ -index of 12 according to [NASA ads](#) and >490 citations,  $h$ -index of 13 according to [Web of Science](#). Here are some selected publications:

- [1] [Oleksii Sokoliuk](#). “Explaining JWST star formation history at  $z \sim 17$  by modifying  $\Lambda$ CDM”. In: *Astron. Astrophys.* 699 (2025), A59.
- [2] Fabiano F. Santos (including [Oleksii Sokoliuk](#)) et al. “Holographic boundary conformal field theory within Horndeski gravity”. In: *JHEP* 12 (2025), p. 217.
- [3] [Oleksii Sokoliuk](#) et al. “AdS Black Hole Thermodynamics and Microstructures from  $f(Q)$  Gravitation”. In: *Fortschritte der Physik* 72.1 (2024), p. 2300043.
- [4] Sanjay Mandal, [Oleksii Sokoliuk](#), et al. “ $H_0$  tension in torsion-based modified gravity”. In: *Nucl. Phys. B* 993 (2023), p. 116285.
- [5] [Oleksii Sokoliuk](#), Simran Arora, et al. “On the impact of  $f(Q)$  gravity on the large scale structure”. In: *Mon. Not. Roy. Astron. Soc.* 522.1 (2023), pp. 252–267.
- [6] [Oleksii Sokoliuk](#), Alexander Baransky, and P. K. Sahoo. “Compact stars admitting Finch-Skea symmetry in the presence of various matter fields”. In: *Chin. Phys. C* 47.1 (2023), p. 015104.
- [7] Fabiano F. Santos, [Oleksii Sokoliuk](#), and Alexander Baransky. “Holographic Complexity of Brane-world in Horndeski Gravity”. In: *Fortschritte der Physik* 71.2-3 (2023), p. 2200141.
- [8] Fabiano F. Santos (including [Oleksii Sokoliuk](#)) et al. “AdS/BCFT Correspondence and Horndeski Gravity in the Presence of Gauge Fields: Holographic Paramagnetism/ Ferromagnetism Phase Transition”. In: *Fortschritte der Physik* 71.12 (2023), p. 2300008.
- [9] [Oleksii Sokoliuk](#) and Alexander Baransky. “Cosmological constraints on bulk viscous  $f(Q, T)$  gravity”. In: *Astron. Nachr.* 343.5 (2022), e220003.
- [10] [Oleksii Sokoliuk](#), Alexander Baransky, Andrew Khorolskiy, et al. “An X-Ray and Optical Study of the UGSU-Type Dwarf Nova Gaia18awg”. In: *Journal of Physical Studies* 26.3 (Sept. 2022), pp. 3901–3909.
- [11] [Oleksii Sokoliuk](#), Alexander Baransky, and P. K. Sahoo. “Kuchowicz gravastars in the braneworld formalism”. In: *Phys. Lett. B* 829 (2022), p. 137048.
- [12] [Oleksii Sokoliuk](#), Alexander Baransky, and P. K. Sahoo. “Probing the existence of the ZTF Casimir wormholes in the framework of  $f(R)$  gravity”. In: *Nucl. Phys. B* 980 (2022), p. 115845.
- [13] [Oleksii Sokoliuk](#), Alexander Baransky, and Pradyumn Kumar Sahoo. “Non-singular T-K axion stars with/without the dynamical bosonic field in the presence of negative  $\Lambda$  term”. In: *Phys. Dark Univ.* 35 (2022), p. 100972.
- [14] [Oleksii Sokoliuk](#), Zinnat Hassan, et al. “Traversable wormholes with charge and non-commutative geometry in the  $f(Q)$  gravity”. In: *Annals Phys.* 443 (2022), p. 168968.
- [15] [Oleksii Sokoliuk](#), Sanjay Mandal, et al. “Generalised Ellis–Bronnikov wormholes in  $f(R)$  gravity”. In: *Eur. Phys. J. C* 82.4 (2022), p. 280.
- [16] [Oleksii Sokoliuk](#), Sneha Pradhan, et al. “Buchdahl quark stars within  $f(Q)$  theory”. In: *Eur. Phys. J. Plus* 137.9 (2022), p. 1077.
- [17] [Oleksii Sokoliuk](#), Subhrat Praharaj, et al. “Accretion flows around exotic tidal wormholes - I. Ray-tracing”. In: *Astron. Astrophys.* 665 (2022), A139.
- [18] [Oleksii Sokoliuk](#) and Alexander Baransky. “On the existence and stability of traversable wormhole solutions in modified theories of gravity”. In: *Eur. Phys. J. C* 81.8 (2021), p. 781.

Observations of near-earth objects and comets were published in 23 Minor Planet Electronic Circulars (MPEC ) and observations of Gamma Ray Bursts in 2 GRB Coordinates Network circulars (GCN )

## COLLABORATIVE PAPERS

Additionally, within several collaborations, I have actively contributed to various publications, engaging in tasks such as observations, data analysis, figure/table creation, and text revision:

- [1] Eleonora Di Valentino (including [Oleksii Sokoliuk](#)) et al. “The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics”. In: *Phys. Dark Univ.* (Apr. 2025).
- [2] T Hussenot-Desenonges (including [Oleksii Sokoliuk](#)) et al. “Multi-band analyses of the bright GRB 230812B and the associated SN2023pel”. In: *Mon. Not. Roy. Astron. Soc.* (2024), stae503.

- [3] Jialian Liu (including [Oleksii Sokoliuk](#)) et al. “Early-time Observations of SN 2023wrk: A Luminous Type Ia Supernova with Significant Unburned Carbon in the Outer Ejecta”. In: *Astrophys. J.* **973.2** (2024), p. 117.
- [4] I. Tosta e Melo (including [Oleksii Sokoliuk](#)) et al. “Ready for O4 II: GRANDMA observations of Swift GRBs over eight weeks in spring 2022”. In: *Astron. Astrophys.* **682** (2024), A141.
- [5] D. A. Kann (including [Oleksii Sokoliuk](#)) et al. “GRANDMA and HXMT Observations of GRB 221009A: The Standard Luminosity Afterglow of a Hyperluminous Gamma-Ray Burst—In Gedenken an David Alexander Kann”. In: *Astrophys. J. Lett.* **948.2** (2023), p. L12.
- [6] V. Aivazyan (including [Oleksii Sokoliuk](#)) et al. “GRANDMA observations of ZTF/Fink transients during summer 2021”. In: *Mon. Not. Roy. Astron. Soc.* **515.4** (2022), pp. 6007–6022.

## CONFERENCE PROCEEDINGS & OTHER

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- [1] Cristina Andrade (including [Oleksii Sokoliuk](#)) et al. “GRANDMA Observations of SN 2023wrk, a Luminous Type Ia Supernova with Significant Unburned Carbon in the Outer Ejecta”. In: *Research Notes of the AAS* **8.10** (Oct. 2024), p. 273.
- [2] S. Agayeva (including [Oleksii Sokoliuk](#)) et al. “The GRANDMA network in preparation for the fourth gravitational-wave observing run”. In: *Observatory Operations: Strategies, Processes, and Systems IX* **12186** (Aug. 2022), 121861H.

## IN PREPARATION & UNDER REVIEW

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- [1] [Oleksii Sokoliuk](#) et al. “The cosmic web’s Lyman- $\alpha$  glow at  $z \approx 2.5$ ; varying hydrodynamic models, dust, and wide-field, narrow-band imaging detection”. In: *arXiv: 2510.07259* (Oct. 2025).
- [2] Kenneth M. Lanzetta (including [Oleksii Sokoliuk](#)) et al. “Direct Images of the Cosmic Web of Intergalactic and Circumgalactic Gas in the Distant Universe”. In: *arXiv: 2412.10081* (Dec. 2024).

## PEER REVIEW

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I have been invited as a peer reviewer for the following journals (most of the reviews are linked to my [ORCID](#) and [Web of Science](#) profiles):

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>➤ European Physical Journal C (2022)</li> <li>➤ New Astronomy (2022)</li> <li>➤ Foundations of Physics (2022, 2024)</li> <li>➤ Scientific Reports (2023)</li> <li>➤ Annalen der Physik (2023)</li> <li>➤ Indian Journal of Physics (2023)</li> <li>➤ Physics of the Dark Universe (2023 (2))</li> </ul> | <ul style="list-style-type: none"> <li>➤ Pramana (2024)</li> <li>➤ Annals of Physics (2024 (3))</li> <li>➤ Physics Letters B (2024)</li> <li>➤ Int. J. Mod. Phys. A (2024)</li> <li>➤ General Relativity and Gravitation (2025)</li> <li>➤ Nuclear Physics B (2025)</li> <li>➤ National Science Review (2025)</li> </ul> |
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## MEMBERSHIPS

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- International Society for Relativistic Quantum Information (ISRQI)
- International Society for Quantum Gravity (ISQG)
- Global Rapid Advanced Network Devoted to the Multi-messenger Addicts (GRANDMA)
- Wide-field Spectroscopic Telescope (WST) Science Team
- Condor Array Telescope
- Assembling Galaxies Of Resolved Anatomy (AGORA)
- Nucleosynthesis Grid (NuGrid)
- Insitute of Physics (IoP) Associate Member
- American Astronomical Society (AAS) Undergraduate Student Member

## AWARDS & GRANTS

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### Recognitions:

- ◆ Nominated for the "Cambridge Independent Science and Technology Awards" by Prof. John Webb

### Monetary awards - 11000\$ total:

- ◆ CA21106 STSM Grant - 2100\$
- ◆ CA21136 ITC Conference Grant for CCP2023 - 2200\$
- ◆ CA21136 STSM Grant - 2200\$
- ◆ CA21136 Conference Grant for CosmoVerse@Lisbon - 1000\$
- ◆ CA21136 Conference Grant for CosmoVerse@Krakow - 1000\$ (declined due to personal reasons)
- ◆ Scholarship of the President of Ukraine - 1000\$
- ◆ MAO NAS travel grant - 3×500\$

### Computing time allocations - 0.85M CPUh total:

- ◆ Co-PI: 300k CPUh on IUCAA Pegasus, with Prof. Jiajun Zhang as Co-PI and Prof. P. K. Sahoo as PI
- ◆ Co-PI: ~200k CPUh on Australian NCI Gadi with Subhrat Praharaj as PI
- ◆ Co-PI: ~300k CPUh on IUCAA Pegasus with Prof. P. K. Sahoo as PI
- ◆ Co-PI: ~50k CPUh on OzSTAR with Prof. John Webb as PI

### Observational proposals - 21ks total:

- ◆ PI: Target of Opportunity (TOO) observation of Gaia18awg by SWIFT space telescope (ID: 13502, ~3k seconds)
- ◆ PI: Target of Opportunity (TOO) observation of Gaia18awg by XMM-Newton space telescope (ID: 08711910011, ~18k seconds)

## PUBLIC OUTREACH

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- ☐ Short article "*Is Modified Gravity an Illusion?*" for CosmoVerse COST action in a series of articles for general public "Learn about Cosmology"
- ☐ Interview for CosmoVerse COST action in a "*Meet our scientists*" dissemination activity
- ☐ Volunteer for Astronomy night under "*Curiosity in Action*" program (funded by STFC)
- ☐ STEM Ambassador Scotland, volunteering to promote science to the general public
- ☐ Demonstrator for the STEM Summer Showcase 2024 at UoA
- ☐ Keynote speaker for ~250 people of general public at the event "*Stargazing in the Botanic Gardens*"

## SKILLS

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|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LANGUAGES          | Ukrainian (native tongue), Russian (native tongue), English (~ C2)                                                                                            |
| PROGRAMMING        | Python (advanced), $\text{\LaTeX}$ (advanced), Mathematica (advanced), MATLAB (intermediate), Haskell (intermediate), Futhark/ML (beginner), C/C++ (beginner) |
| PARALLEL COMPUTING | MPI, OpenMP, Slurm, worked with Cray/ARM                                                                                                                      |

## PROJECTS

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|---------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>CA21106 - COSMIC WISPerS in the Dark Universe</b><br>– Working Group 2: WISPerS Dark Matter and Cosmology                          | Oct 2022 – Oct 2026 |
| <b>CA21136 - CosmoVerse</b><br>– Working Group 3: Fundamental Physics                                                                 | Oct 2022 – Oct 2026 |
| <b>CA22113 - Fundamental challenges in theoretical physics</b><br>– Working Group 2: Gravity and Holography                           | Oct 2023 – Oct 2027 |
| <b>Mathematical Modeling in Interdisciplinary Research of Processes and Systems</b><br>– Grant for the Lab of LSS, MAO NAS of Ukraine | Jan 2021 – Jan 2025 |

## DEVELOPED SOFTWARE & SIMULATIONS

|                        |                                                                                                                                                                                                                                                                                                                                              |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RAMA                   | Code can be used with <code>reps</code> output to find $z_*$ , $l$ at which N-body simulations coincide with the desired cosmology in the presence of massive neutrinos. Available at <a href="#">RAMA</a> .                                                                                                                                 |
| MG-SWIFT               | Modification of the standard N-body/SPH SWIFT code to include such cosmologies as $f(Q)/f(T)$ gravitation, Palatini- $f(R)$ gravitation, with the help of linearly interpolated Hubble parameter and effective gravitational constant, which are updated every step using provided data from tables. Available at <a href="#">MG-SWIFT</a> . |
| HASKY                  | Solver for the heavy metal isotope abundances that uses countless sources for nucleosynthesis yields, such as AGB/S-AGB, CCSN/HNe/MRSNe and SN Ia, Novae, WR stars and more, developed with the type safety of Haskell. Available at <a href="#">HASKY</a> .                                                                                 |
| LANCELOT               | A suite of 12 high-resolution N-body simulations with more than 13.5 billion particles in total as well as 24 TB of output. From the simulation snapshots, many quantities such as void/halo catalogues, power spectrum, and HMF are being derived. Currently in development.                                                                |
| CORINTH                | Emulator of non-linear $P(k)$ for Jordan–Brans–Dicke cosmology, based on 200 Comoving Lagrangian Approximation simulations of structure formation with Latin hypercube parameter sampling. Currently in development.                                                                                                                         |
| ANTHOLOGY OF INFLATION | A project aimed at studying 27 single-parameter models of inflation using $N = 256$ lattice simulations of reheating, primordial power spectra of curvature perturbations, and gravitational waves. Currently in development; some of the code is available via <a href="#">ASPIC</a> .                                                      |

## REFERENCES

|              |                                                                                              |              |                                                            |
|--------------|----------------------------------------------------------------------------------------------|--------------|------------------------------------------------------------|
| NAME:        | <b>Dr. P. K. Sahoo</b>                                                                       | NAME:        | <b>Dr. Irina Vavilova</b>                                  |
| POSITION:    | Professor                                                                                    | POSITION:    | Professor                                                  |
| AFFILIATION: | Dept. of Mathematics,<br>BITS Pilani                                                         | AFFILIATION: | MAO,<br>NAS of Ukraine                                     |
| E-MAIL:      | <a href="mailto:pksahoo@hyderabad.bits-pilani.ac.in">pksahoo@hyderabad.bits-pilani.ac.in</a> | E-MAIL:      | <a href="mailto:irivav@mao.kiev.ua">irivav@mao.kiev.ua</a> |
| NAME:        | <b>Dr. Charles Wang</b>                                                                      | NAME:        | <b>Dr. Alexander Baransky</b>                              |
| POSITION:    | Professor                                                                                    | POSITION:    | Senior Scholar                                             |
| AFFILIATION: | Dept. of Physics,<br>University of Aberdeen                                                  | AFFILIATION: | Lisnyky Obs. Station,<br>Kyiv Astron. Obs.                 |
| E-MAIL:      | <a href="mailto:c.wang@abdn.ac.uk">c.wang@abdn.ac.uk</a>                                     | E-MAIL:      | <a href="mailto:abaransky@ukr.net">abaransky@ukr.net</a>   |

## RESEARCH INTERESTS

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I am a cosmologist, mainly interested in the complicated, high resolution simulations of the Large Scale Structure of the Universe within the non-standard theories of modified gravitation or dark matter. As well, I am trying to alleviate  $H_0$ ,  $\sigma_8$  and other cosmological tensions beyond  $\Lambda$ CDM, find new physics in higher order weak lensing statistics.